

Parabolic Inverse Source Reconstruction: From Raw Error to Visible Geometry

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1 Background and Motivation

Recovering the location and shape of a source from a diffused measurement is difficult because diffusion hides fine spatial detail. This project studies a stationary source $f^\dagger$ in the parabolic equation

$$\begin{cases} \partial_t u + Lu = f^\dagger, & (x, t) \in \Omega \times (0, T), \\ u = 0, & (x, t) \in \partial\Omega \times (0, T), \\ u(\cdot, 0) = 0, \end{cases}$$

where L is a positive elliptic operator and only the final state $u(\cdot, T)$ is observed. High-frequency information describing sharp source boundaries is suppressed before reaching the measurement. Attempting to recover it amplifies noise, making the inverse problem ill posed.

My starting point was proper orthogonal decomposition (POD), which constructs a low-dimensional state space from solution snapshots. Previous work shows how this can accelerate the repeated forward PDE solves needed for inverse reconstruction [1, 3]. During numerical investigation, I observed that a reconstructed discontinuous source could have a substantial relative L^2 error while retaining its coarse location and recognizable shape. This led me to ask whether a reconstruction can be inaccurate pixel by pixel yet recover geometrically meaningful information at a prescribed physical scale.

2 Scale-Resolved Reconstruction

The main idea is to distinguish three aspects of reconstruction quality. Raw source error, $\|f_{\text{rec}} - f^\dagger\|_{L^2}$, measures mismatch in source values. To examine what remains visible at a physical length $\ell > 0$, I use the heat operator

$$H_\ell = e^{-\ell^2 L}.$$

Applying H_ℓ to both sources suppresses features much smaller than ℓ and allows their larger-scale structure to be compared. This is smoothing of the source itself, rather than a comparison of final-time measurements. Visible L^2 error measures average field mismatch, while visible L^∞ error measures the largest pointwise discrepancy.

Geometry is then assessed by comparing the level sets of $H_\ell f_{\text{rec}}$ and $H_\ell f^\dagger$ at a chosen threshold τ . Their Hausdorff distance measures the greatest distance from one interface to its nearest point on the other. Source amplitudes, visible fields, and visible boundaries answer different questions: a reconstruction may describe where a source lies more accurately than its values near a discontinuous edge.

For the analysis, the final-time map is $S_T = L^{-1}(I - e^{-TL})$. Tikhonov regularization balances fitting noisy data against a penalty on the source norm [2]. To isolate spatial resolution, I use

spectral truncation as an analytical benchmark: P_r retains the first r eigenmodes of L , rather than a learned POD basis. For data $M^\delta = S_T f^\dagger + \xi$, with $\|\xi\|_{L^2} \leq \delta$, the reconstruction is

$$f_{\lambda,r}^\delta = \arg \min_{f \in L^2(\Omega)} \left\{ \|P_r S_T f - M^\delta\|_{L^2}^2 + \lambda \|f\|_{L^2}^2 \right\}, \quad \lambda > 0.$$

The general analysis assumes a positive self-adjoint operator with compact inverse, prescribed eigenvalue growth, and Gaussian heat-kernel bounds. The Dirichlet Laplacian gives explicit rates for indicator sources.

3 Main Results and Numerical Experiments

For $L = -\Delta$ in two dimensions and an indicator source $f^\dagger = \mathbf{1}_E$, where E is a Lipschitz set lying strictly inside Ω , the raw reconstruction error satisfies, for every $\varepsilon > 0$,

$$\|f_{\lambda,r}^\delta - f^\dagger\|_{L^2} \lesssim \frac{\delta}{\sqrt{\lambda}} + \lambda^{1/8-\varepsilon} + r^{-1/4+2\varepsilon}.$$

The three terms describe noise amplification, regularization bias, and loss from finite spectral resolution. A rectangular indicator gives lower bounds with powers $\lambda^{1/8}$ and $r^{-1/4}$. Thus the slow algebraic behaviour is inherent over this source class, up to the arbitrarily small exponent loss in the upper estimate.

At a fixed visibility scale, the behaviour changes. In two dimensions, for exact data, $0 < \ell \leq \sqrt{T}$, and ℓ above the conservative regularization transition, the visible field satisfies

$$\|H_\ell(f_{\lambda,r}^0 - f^\dagger)\|_{L^\infty} \leq C_{\ell,T,f^\dagger} \left(\lambda + e^{-c\ell^2 r} \right).$$

The regularization factor scales roughly as λ/ℓ^4 , with an additional scale-dependent factor when estimating the pointwise norm. Heat smoothing makes the unresolved rank tail exponentially small, although regularization and noise still limit overall accuracy. The constants also worsen as ℓ decreases: finer visibility demands more accurate reconstruction.

I tested this separation using an exact-data spectral-truncation experiment with a rectangular indicator. The fitted raw-error slope against rank was approximately -0.242 , close to the theoretical $-1/4$ benchmark. Meanwhile, relative visible L^2 error fell from approximately 2.14×10^{-1} to 1.10×10^{-6} , a reduction of more than five orders of magnitude. The experiment illustrates how slow recovery of a discontinuous source can coexist with much faster recovery of its visible field.

To connect field accuracy with geometry, let $F = H_\ell f^\dagger$ and $\widehat{F} = H_\ell f_{\text{rec}}$, with threshold interfaces Γ and $\widehat{\Gamma}$. Suppose Γ is compact and C^1 , $\widehat{F}$ is continuous, and the true field has a signed transverse crossing margin $m > 0$ in a valid tube of width a , with separation from the threshold outside that tube. Then

$$\|\widehat{F} - F\|_{L^\infty} \leq \epsilon < ma \implies d_H(\widehat{\Gamma}, \Gamma) \leq \frac{\epsilon}{m}.$$

A sufficiently steep threshold crossing prevents a small field perturbation from moving the boundary very far. Relative to the original source boundary, smooth interior half-level interfaces have $O(\ell^2)$ smoothing displacement under suitable curvature, reach, and local heat-kernel assumptions; conical points generally exhibit $O(\ell)$ scaling.

A larger-hole block-letter A tested nonconvex visible geometry. Although its relative raw error was approximately 0.417, its relative visible L^2 error was only 0.040. The pointwise visible discrepancy was approximately 0.051, and the sampled boundary distance was 0.0082. A numerical crossing margin of 3.09 gave a displacement estimate of approximately 0.0166, consistent with the smaller measured gap. These numerical values are consistent with the level-set

stability estimate, although the theorem assumes continuous conditions in a neighborhood of the boundary that are only sampled numerically here. Since the letter contains corners, we use the general visible-boundary stability result rather than the $O(\ell^2)$ asymptotic result for smooth interfaces.

4 Conclusion and Significance

Raw L^2 error remains important, but it does not fully describe the geometric information in a reconstruction. Fine boundary oscillations can contribute substantially to source error even when the source is well localized at a coarser scale. The analysis explains this distinction by connecting regularization and spectral resolution to heat-visible fields, then relating pointwise field accuracy to stable interfaces. This helps interpret the recognizable shapes encountered in reduced-order inverse reconstruction: usefulness depends on the physical scale and information of interest, as well as pixelwise accuracy.

References

- [1] Z. Wang, W. Zhang, and Z. Zhang. A data-driven model reduction method for parabolic inverse source problems and its convergence analysis. *Journal of Computational Physics*, 487 (2023), 112156.
- [2] H. W. Engl, M. Hanke, and A. Neubauer. *Regularization of Inverse Problems*. Kluwer, 1996.
- [3] Hu, Zhang, and Zhu. Efficient identification of geometric inverse sources of parabolic problems by model order reduction. *Inverse Problems*, 41 (2025), 055015.